

THREAT DETECTION

SIEM

Enterprise SOC Lab

Security Monitoring, Threat Hunting, and SIEM Operations

VS



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Executive Summary

The Enterprise SOC Lab project was designed to simulate a real-world Security Operations Center (SOC) environment capable of collecting, transporting, indexing, and analyzing endpoint security telemetry from multiple Windows systems.

The lab was built entirely within VMware Workstation Pro and consists of a Windows Server 2025 Active Directory Domain Controller, multiple Windows 11 endpoints, an Ubuntu-based Splunk Enterprise SIEM server, Sysmon endpoint monitoring, and Splunk Universal Forwarders.

The primary objective was to gain hands-on experience with enterprise security monitoring, centralized logging, endpoint telemetry collection, Active Directory administration, SIEM operations, threat hunting, and troubleshooting security monitoring pipelines.

By the completion of the project, all systems were successfully forwarding Sysmon and Windows event data into Splunk Enterprise, enabling centralized visibility and threat hunting across the environment.

Project Goals

The project was created to achieve the following objectives:

- Build a realistic enterprise SOC environment
- Deploy Active Directory infrastructure
- Centralize Windows security logging
- Implement endpoint monitoring using Sysmon
- Deploy Splunk Enterprise as a SIEM platform
- Configure Universal Forwarders
- Create a security monitoring pipeline
- Validate end-to-end log collection
- Perform threat hunting activities
- Develop SOC analyst skills

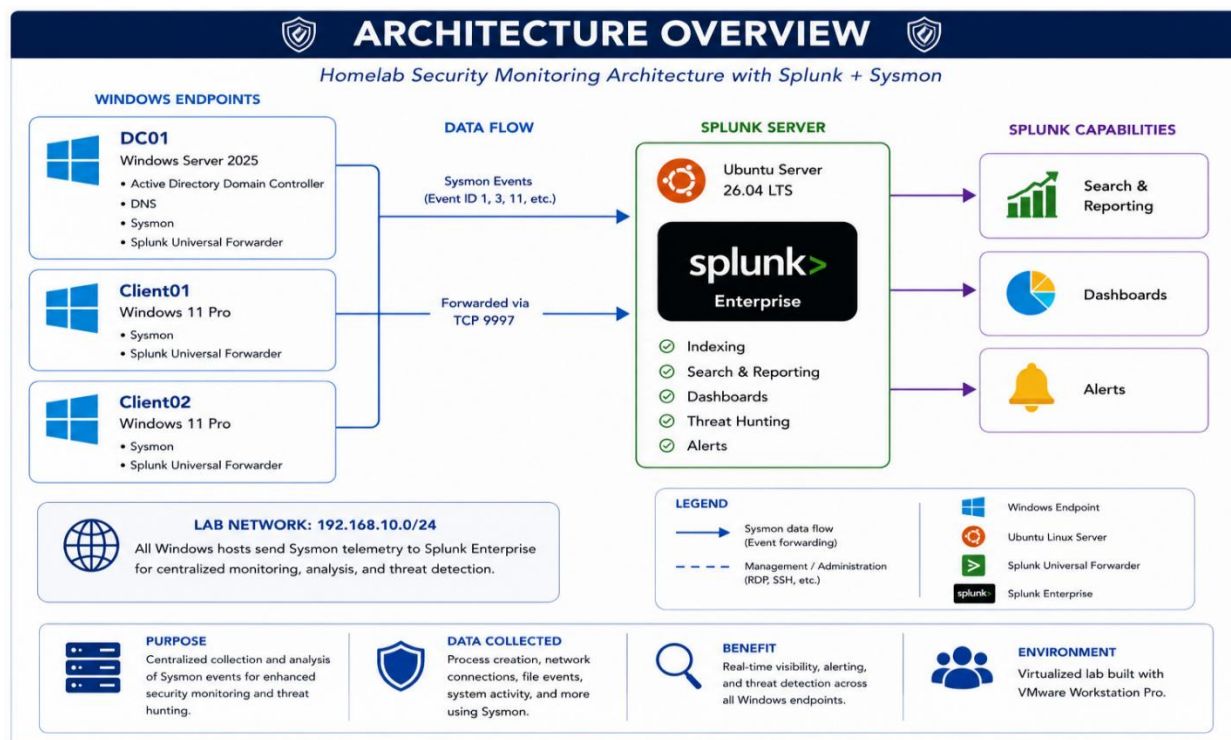
Architecture Overview

The Enterprise SOC Lab was designed to simulate a small enterprise network environment and demonstrate centralized log collection, endpoint telemetry monitoring, and threat hunting capabilities.

The environment consisted of a Windows Server domain controller, two Windows client workstations, and a dedicated Ubuntu-based Splunk Enterprise server. Sysmon was

deployed to all Windows systems to generate enhanced endpoint telemetry, while Splunk Universal Forwarders were configured to securely forward logs to the SIEM platform. This architecture enabled the collection, transportation, indexing, and analysis of security events from multiple endpoints, providing visibility into process execution, system activity, and potential security threats.

Figure 1 – Enterprise SOC Lab Architecture Diagram



Virtual Infrastructure

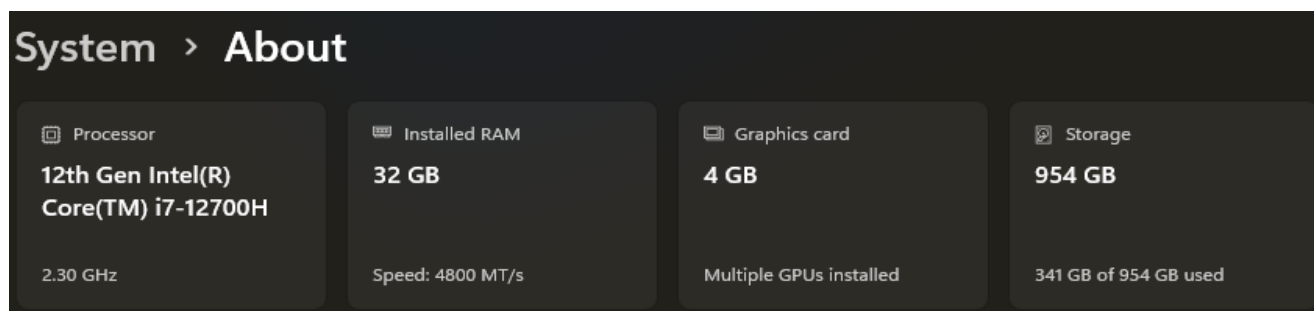
Hypervisor

VMware Workstation Pro

Host System

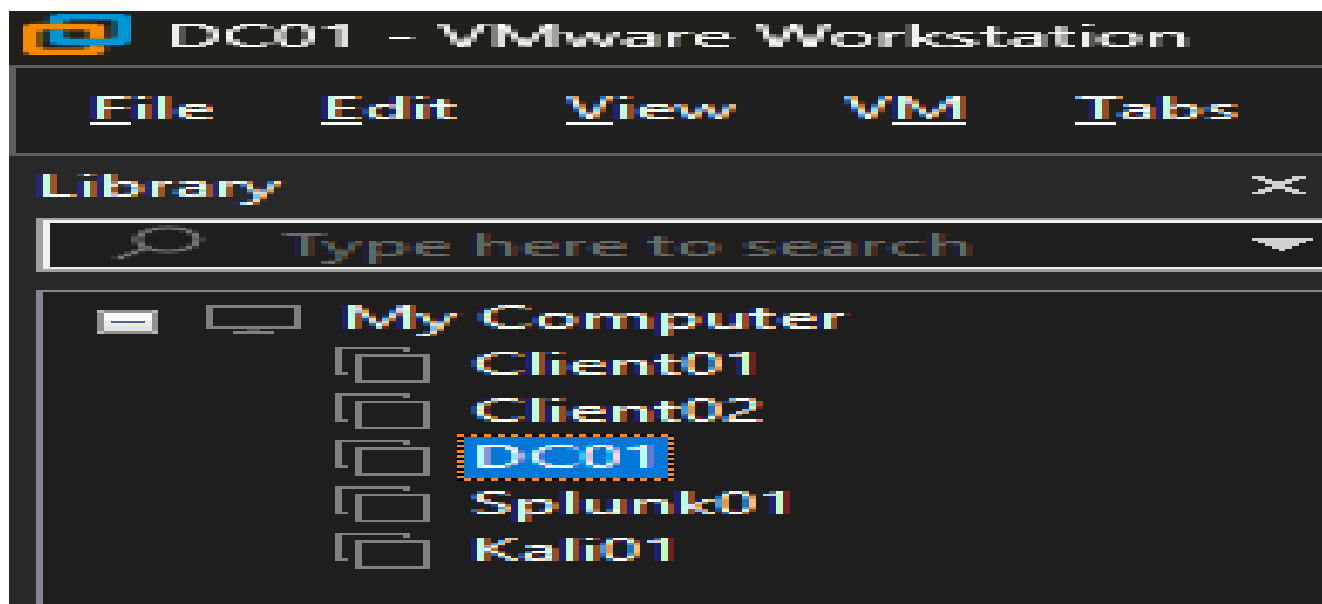
- Windows 11
- 32 GB RAM
- Hardware Virtualization Enabled

Figure 2 – Host System Specifications



Virtual Machines

Figure 3 – VMware Workstation Pro Virtual Machines



DC01

Role:

Windows Server 2025 Domain Controller

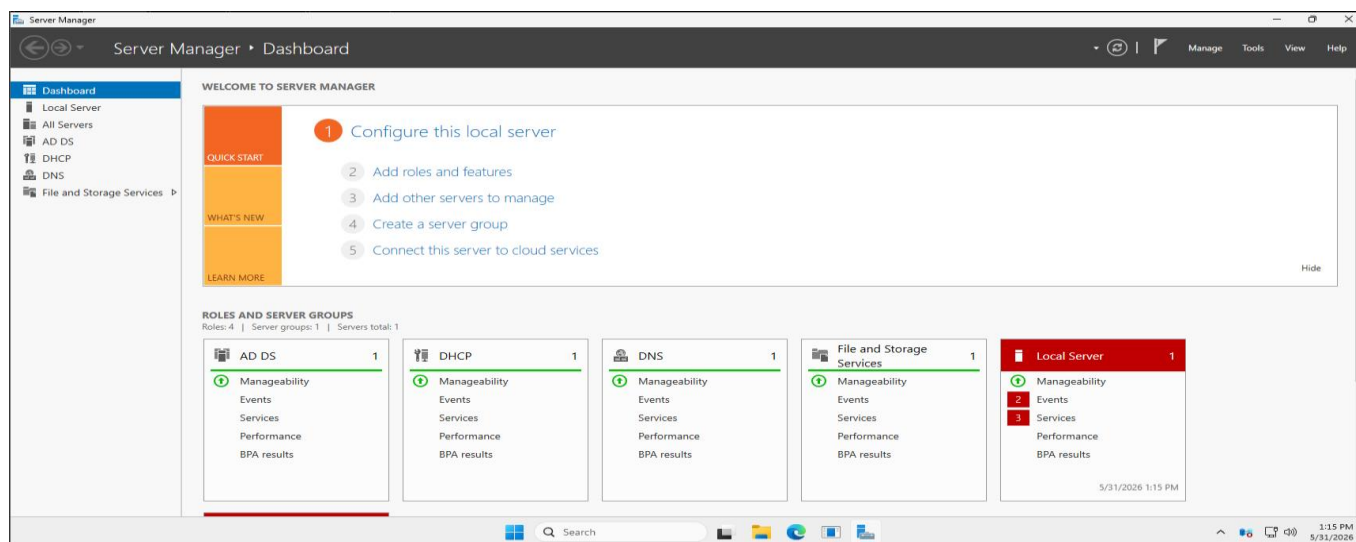
Functions:

- Active Directory
- DNS
- Domain Services
- Centralized Authentication

IP Address:

192.168.10.10

Figure 4 – DC01 Server Manager Dashboard



Splunk01

Role:

Ubuntu Server running Splunk Enterprise

Figure 5 – Splunk Service Running on Ubuntu

```
elie@splunk01:~$ sudo /opt/splunk/bin/splunk status
splunkd is running (PID: 2022).
splunk helpers are running (PIDs: 2023 2155 2249 2287 3272).
```

Functions:

- Log Collection
- Log Indexing
- Threat Hunting
- Security Monitoring

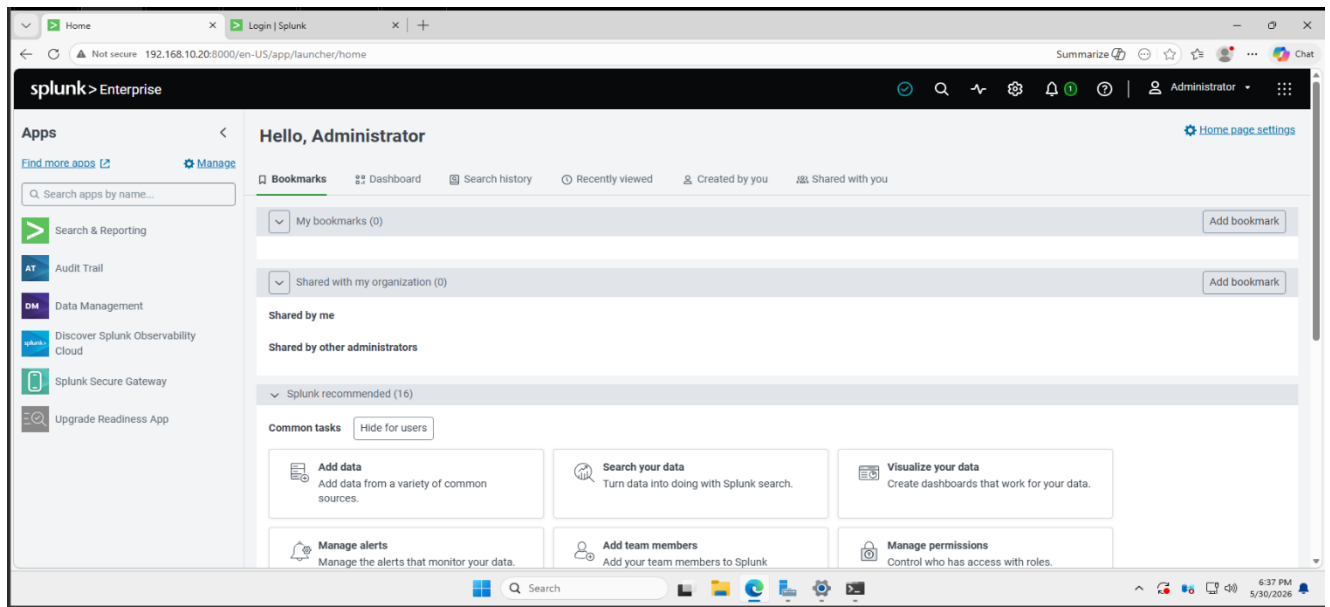
IP Address:

192.168.10.20

Ports:

- 8000/TCP (Web Interface)
- 9997/TCP (Forwarder Ingestion)

Figure 6 – Splunk Enterprise Dashboard



Client01

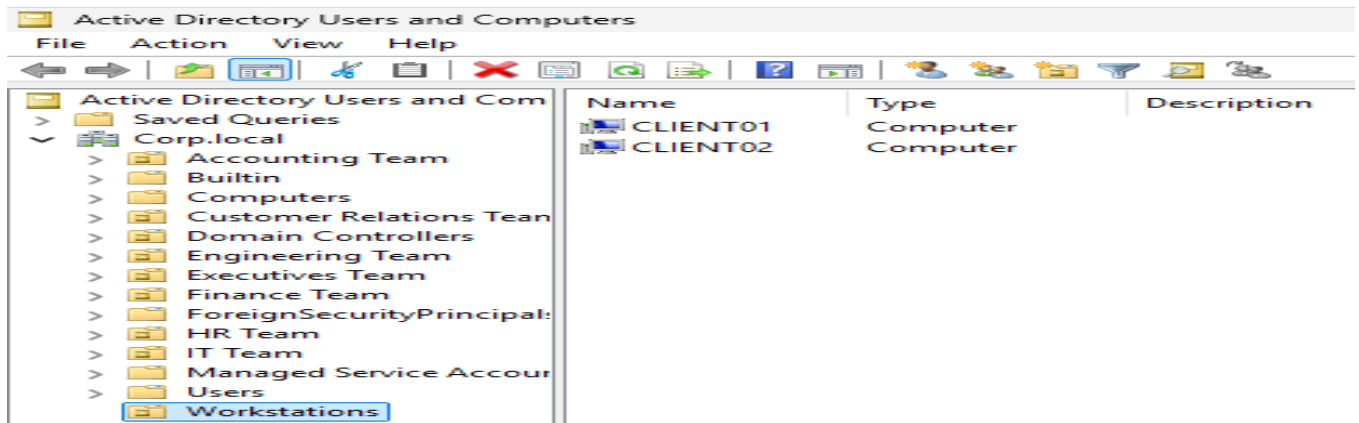
Role:

Windows 11 Endpoint

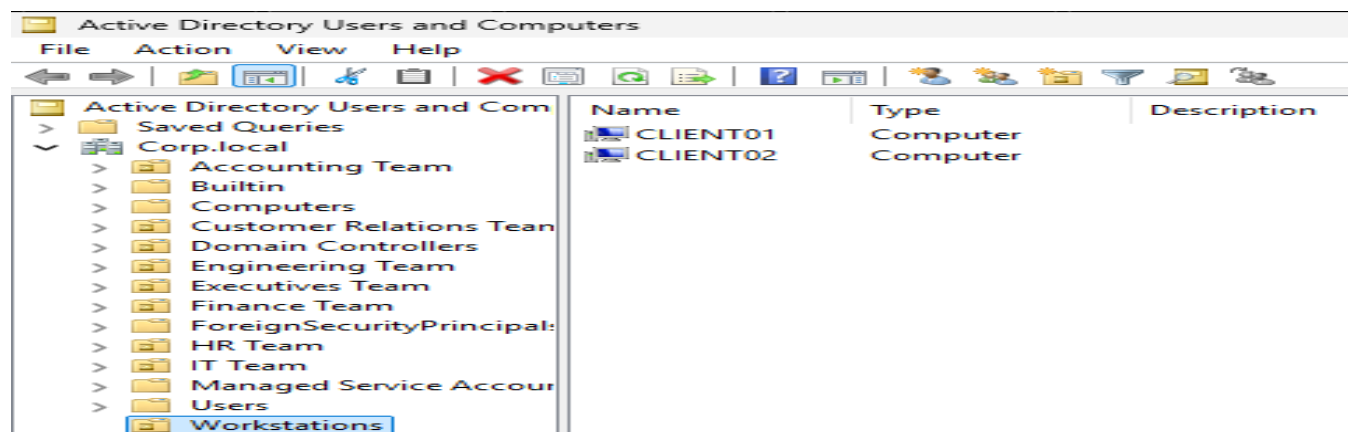
Functions:

- Domain Workstation
- Sysmon Endpoint
- Log Source

Figure 7 – Domain-Joined Client Workstations



Client02



Role:

Windows 11 Endpoint

Functions:

- Domain Workstation
- Sysmon Endpoint
- Log Source

Kali Linux

Role:

Security Testing Platform

Functions:

- Vulnerability Assessment
- Network Scanning
- Attack Simulation

IP Address:

192.168.10.30

Network Architecture

Domain:

corp.local

Internal Network:

192.168.10.0/24

Communication Flow:

DC01 ↔ Client01 ↔ Client02 ↔ Splunk01 ↔ Kali Linux

Tools Used

Figure 8 – Security Monitoring Pipeline Diagram

TOOLS USED			
TOOL	PURPOSE	DESCRIPTION	HOW IT WAS USED
 Splunk Enterprise (Ubuntu Server 22.04 LTS)	 SIEM Platform	Enterprise platform for collecting, indexing, searching, and analyzing machine data for security monitoring and threat hunting.	- Deployed on Ubuntu server - Ingested Sysmon logs from Windows endpoints - Built searches, dashboards, and alerts - Performed threat hunting and analysis
 Sysmon (Windows)	 Windows System Monitoring	Microsoft Sysinternals tool that provides detailed telemetry about system activity, including process creation, network connections, and more.	- Installed on all Windows hosts - Configured to capture Event ID 1, 3, 11, and other relevant events - Forwarded events to Splunk via Universal Forwarder
 Splunk Universal Forwarder (Windows)	 Data Collection & Forwarding	Lightweight agent that collects data from sources and forwards it securely to Splunk Enterprise.	- Installed on DC01, Client01, Client02 - Configured to monitor Sysmon Operational log - Forwarded data to Splunk Enterprise over TCP 9997
 VMware Workstation Pro (Windows)	 Virtualization Platform	Desktop hypervisor used to create and manage virtual machines for the homelab environment.	- Hosted the entire lab environment - Ran Windows Server 2022, Windows 10 clients, and Ubuntu server
 Ubuntu 22.04 LTS (Linux)	 Operating System	Linux distribution used as the operating system for the Splunk Enterprise server.	- Installed Splunk Enterprise - Managed Splunk services - Hosted indexes and data
 PowerShell (Windows)	 Automation / Configuration	Command-line shell and scripting language used for automation, configuration, and system administration.	- Installed Sysmon silently - Verified services and configurations - Checked connectivity (Test-NetConnection) - Managed Windows services and system settings

The following tools and technologies were utilized throughout the project.

Tool

Purpose

VMware Workstation Pro

Virtualization platform used to host all virtual machines

Windows Server 2025

Active Directory Domain Services

Windows 11 Pro

Endpoint workstations

Ubuntu Server

Splunk Enterprise hosting platform

Splunk Enterprise

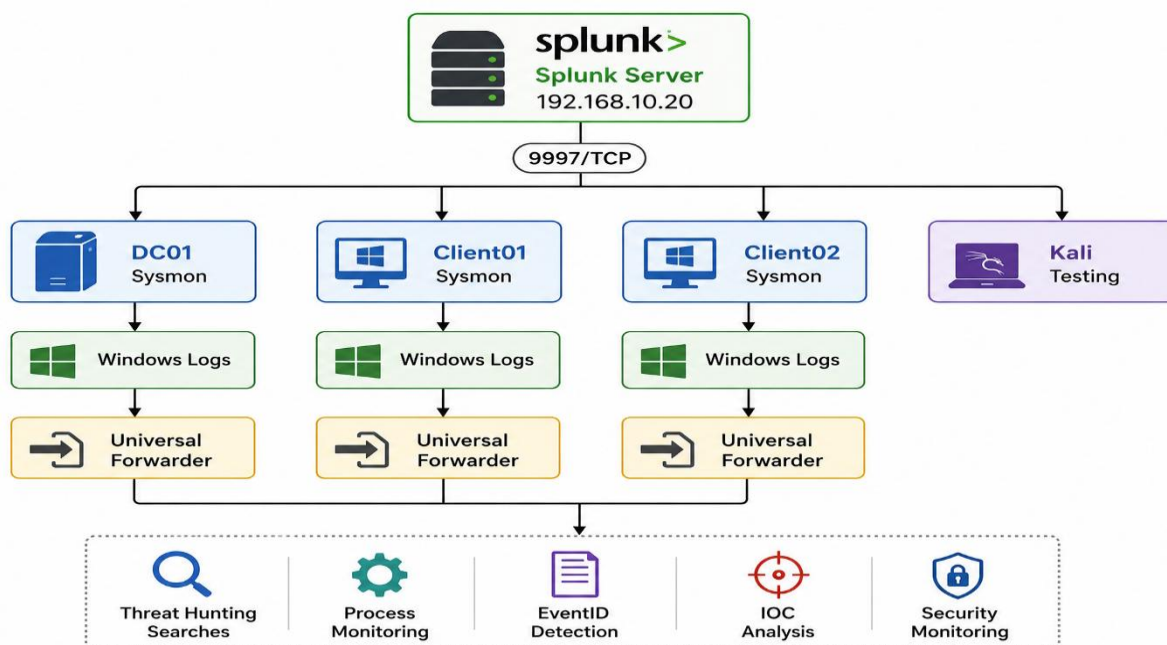
Security Information and Event Management (SIEM)

Splunk Universal Forwarder	Log forwarding agent
Sysmon	Endpoint telemetry collection
SwiftOnSecurity Sysmon Configuration	Advanced event monitoring rules
PowerShell	Administration and troubleshooting
DNS	Name resolution services
Active Directory	Identity and access management
TCP Port 9997	Splunk forwarding communications

These technologies were integrated to create a centralized security monitoring platform capable of collecting, transporting, indexing, and analyzing endpoint telemetry across multiple systems.

Security Monitoring Pipeline

Figure 9 – Security Monitoring Pipeline Diagram



The most important component of the project was the security monitoring pipeline.

This pipeline enabled security events generated on Windows systems to be collected, transported, indexed, and analyzed.

Pipeline Flow:

Windows Endpoint

↓

Sysmon

↓

Windows Event Logs

↓

Splunk Universal Forwarder

↓

TCP Port 9997

↓

Splunk Enterprise

↓

Indexing

↓

Field Extraction

↓

Threat Hunting

↓

Security Investigation

Sysmon Deployment

Sysmon was installed on:

- DC01
- Client01

- Client02

Figure 10 – Sysmon Service Running on DC01

```

Administrator: Windows PowerShell
PS C:\Users\Administrator> hostname
DC01
PS C:\Users\Administrator> Get-Service Sysmon64

Status      Name      DisplayName
-----
Running     Sysmon64  Sysmon64
  
```

Figure 11 – Sysmon Service Running on Client01

```

Administrator: Windows PowerShell
PS C:\Users\Administrator> hostname
Client01
PS C:\Users\Administrator> Get-Service Sysmon64

Status      Name      DisplayName
-----
Running     Sysmon64  Sysmon64
  
```

Figure 12 – Sysmon Service Running on Client02

```

Administrator: Windows PowerShell
PS C:\Users\Administrator> Hostname
Client02
PS C:\Users\Administrator> Get-Service Sysmon64

Status      Name      DisplayName
-----
Running     Sysmon64  Sysmon64
  
```

Configuration Used:

SwiftOnSecurity Sysmon Configuration

Telemetry Generated:

Event ID 1 – Process Creation

Event ID 3 – Network Connections

Event ID 11 – File Creation

Event ID 13 – Registry Modifications

Event ID 22 – DNS Queries

Event Channel:

Microsoft-Windows-Sysmon/Operational

Splunk Enterprise Deployment

Splunk Enterprise was installed on Ubuntu Server.

Configuration Included:

- Search Head
- Indexer
- Receiver

Receiving Port:

9997/TCP

Web Interface:

8000/TCP

Purpose:

Centralized Security Monitoring

Universal Forwarder Deployment

Universal Forwarder was installed on:

- DC01
- Client01
- Client02

Figure 13 – Universal Forwarder Installation

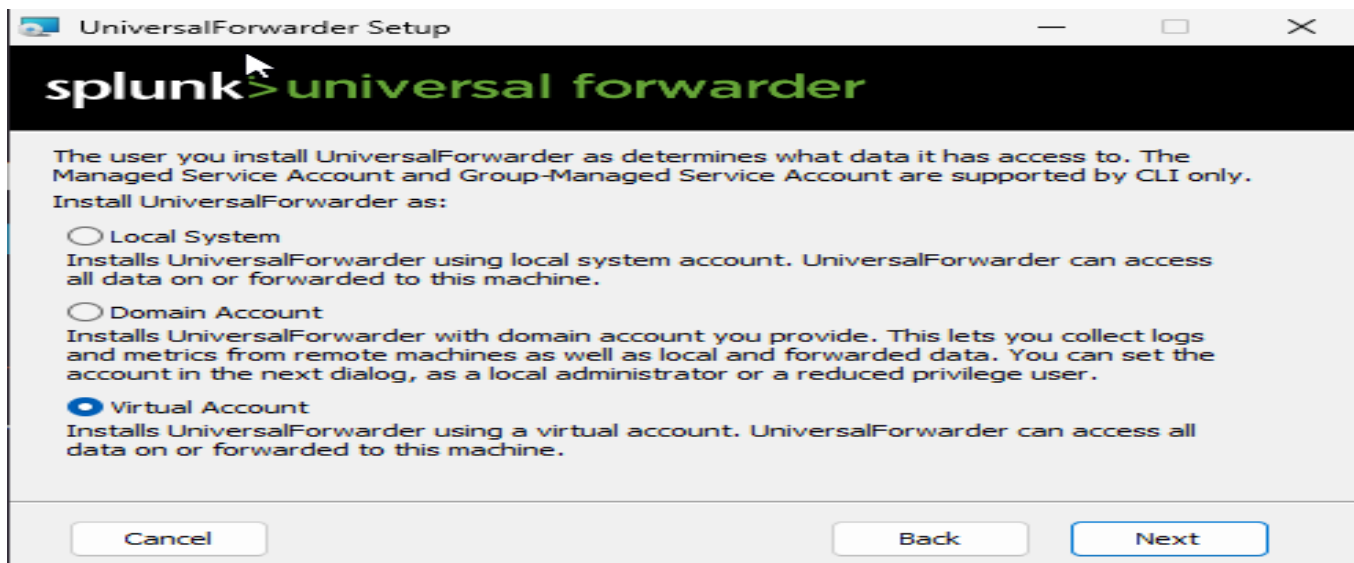


Figure 14 – Splunk Forwarder Service Running

Services (Local)					
SplunkForwarder					
Stop the service Restart the service	Name	Description	Status	Startup Type	Log On As
	SplunkForwarder	SplunkForw...	Running	Automatic	Local System
Description: SplunkForwarder is the remote data collection service for Splunk, a data platform for operational intelligence. If it is stopped, Splunk will stop collecting and sending data to Splunk indexes, which may result in data loss. Please see www.splunk.com for more information. Questions can be submitted to www.splunk.com/answers or for supported customers www.splunk.com/page/submit_issue	Software Protection	Enables the ...		Automatic (D...	Network S...
	Spot Verifier	Verifies pote...		Manual (Trig...	Local Syste...
	SSDP Discovery	Discovers n...		Manual	Local Service
	State Repository Service	Provides re...	Running	Automatic	Local Syste...
	Still Image Acquisition Events	Launches a...		Manual	Local Syste...
	Storage Service	Provides en...	Running	Automatic (D...	Local Syste...
	Storage Tiers Management	Optimizes t...		Manual	Local Syste...
	Sync Host_90ffe	This service ...	Running	Automatic (D...	Local Syste...
	SysMain	Maintains a...	Running	Automatic	Local Syste...
	Sysmon64	System Mo...	Running	Automatic	Local Syste...
	System Event Notification S...	Monitors sy...	Running	Automatic	Local Syste...
	System Events Broker	Coordinates...	Running	Automatic (T...	Local Syste...
	Task Scheduler	Enables a us...	Running	Automatic (T...	Local Syste...
	TCP/IP NetBIOS Helper	Provides su...	Running	Manual (Trig...	Local Service

Purpose:

Collect and forward endpoint telemetry.

Inputs Configured:

- Windows Security Logs
- Windows System Logs
- Windows Application Logs
- Sysmon Operational Logs

Destination:

192.168.10.20:9997

Major Challenges and Troubleshooting

Issue 1 – Clients Not Appearing in Splunk

Figure 15 – Universal Forwarder Connectivity Validation

```

PS C:\Program Files\SplunkUniversalForwarder\bin> .\splunk.exe list forward-server
Active forwards:
    192.168.10.20:9997
Configured but inactive forwards:
    None
PS C:\Program Files\SplunkUniversalForwarder\bin> |
  
```

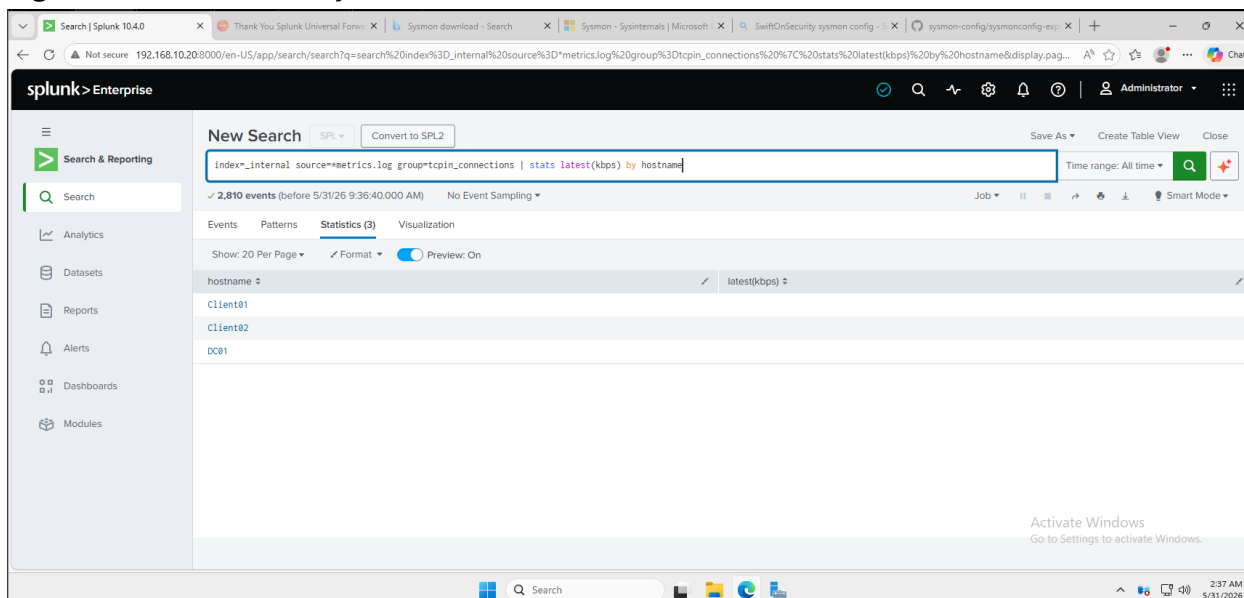
Problem:

Client01 and Client02 were not visible in Splunk searches.

Investigation:

- Tested connectivity
- Verified outputs.conf
- Verified forwarding status
- Confirmed port 9997 access

Figure 16 – Host Visibility Validation



The screenshot shows the Splunk Enterprise web interface. The search bar contains the query: `index=_internal source=metrics.log group=per_host_thruput | stats sum(kb) by series`. The search results show 3,460 events. The 'Statistics (6)' tab is selected, displaying a table with the following data:

series	sum(kb)
\$decisionstartup	0.000
127.0.0.1	59.961
client01	7995.407
client02	6387.467
dc01	142641.751
splunk01	494597.692

Resolution:

Validated Universal Forwarder connectivity and restarted services.

Result:

Clients became visible in Splunk.

Issue 2 – Sysmon Events Missing

Figure 17 – Initial Sysmon Search Returning No Results

The screenshot shows the Splunk Enterprise web interface. The search bar contains the query: `index=* sourcetype="XmlEventLog:Microsoft-Windows-Sysmon/Operational" | stats by host source`. The search results show 0 events. The 'Statistics (0)' tab is selected, displaying the message: "No results found."

Problem:

Sysmon events were not appearing from Client01 and Client02.

Figure 18 – Forwarder Initial Service Account Configuration

```

PS C:\Program Files\SplunkUniversalForwarder\bin> Get-WmiObject Win32_Service | Where-Object {$_.Name -like "*splunk*"} | Select Name, StartName, State

Name                StartName              State
-----                -
SplunkForwarder     NT SERVICE\SplunkForwarder Running
  
```

Error Observed:

Could not subscribe to Windows Event Log channel

Microsoft-Windows-Sysmon/Operational

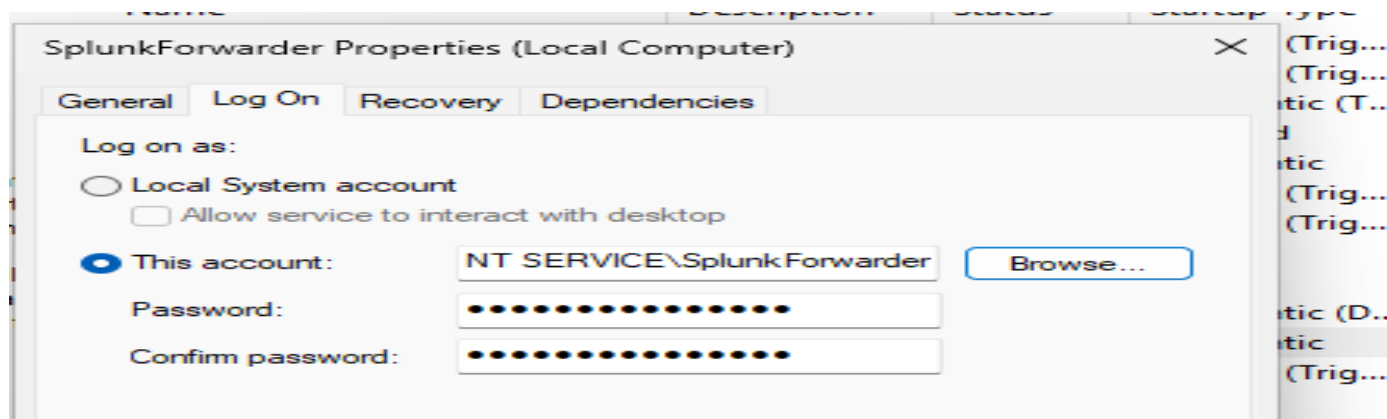
errorCode=5

Root Cause:

The Splunk Forwarder service account lacked permission to read the Sysmon channel.

Original Service Account:

NT SERVICE



Resolution:

Changed service account to:

Local System Account

Figure 19 – Forwarder Updated Service Account Configuration

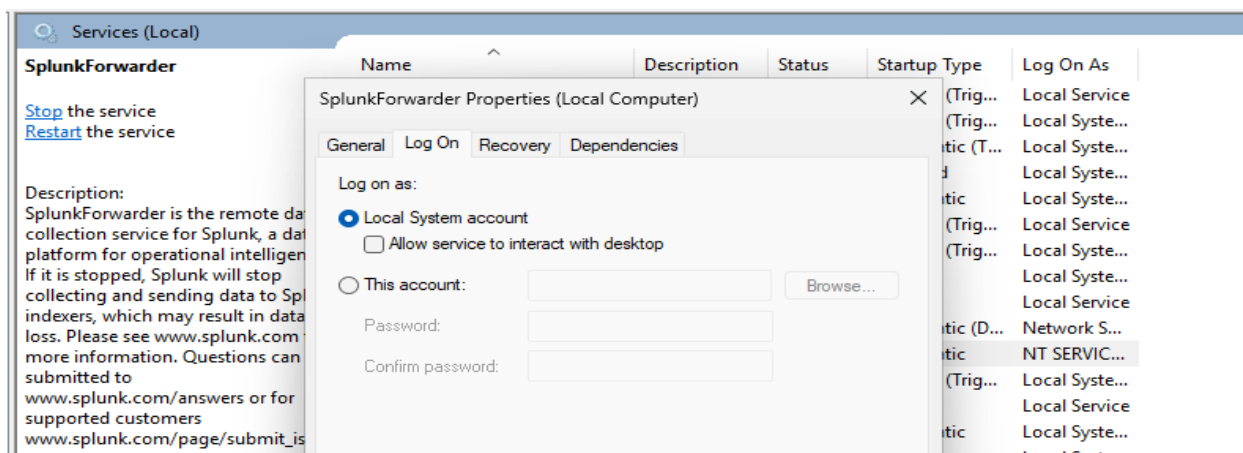
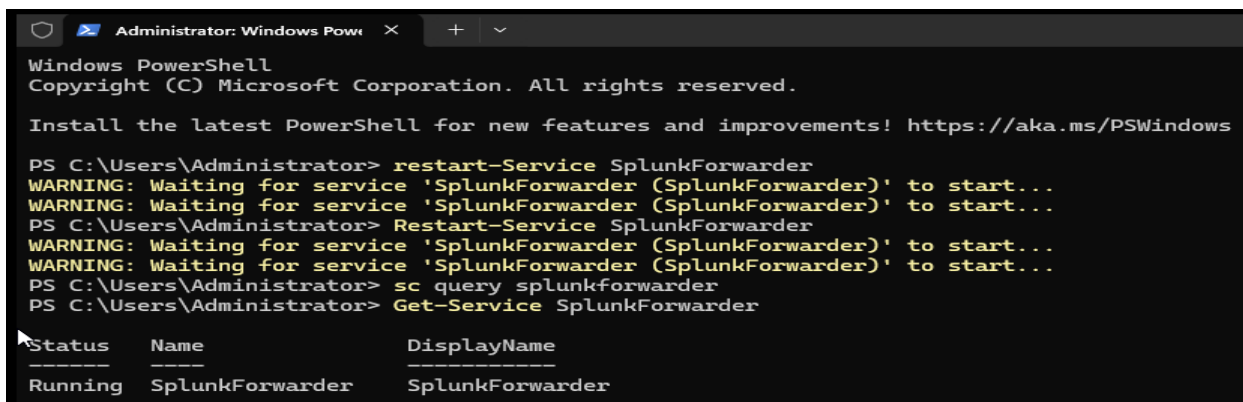


Figure 20 – Restarted Splunk Forwarder service.

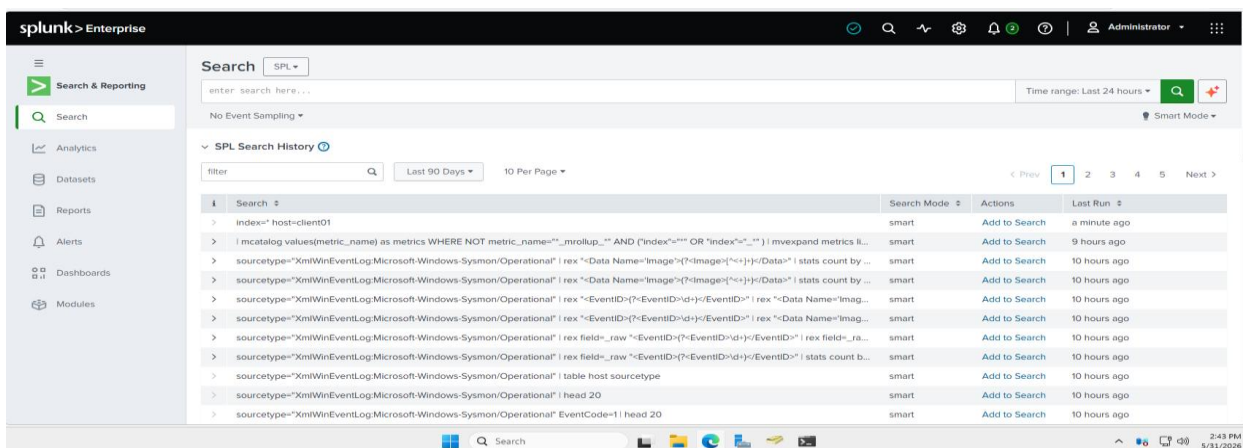


Result:

Sysmon telemetry immediately began flowing into Splunk.

Issue 3 – EventID Searches Returning No Results

Figure 21 – Sysmon Telemetry Successfully Flowing



Problem:

Sysmon events were stored as XML and fields were not extracted.

Resolution:

Created field extractions using Splunk rex commands.

Examples:

EventID Extraction

Image Extraction

CommandLine Extraction

ParentImage Extraction

Result:

Security events became searchable and useful for threat hunting.

Validation Testing

Host Visibility Validation

Search:

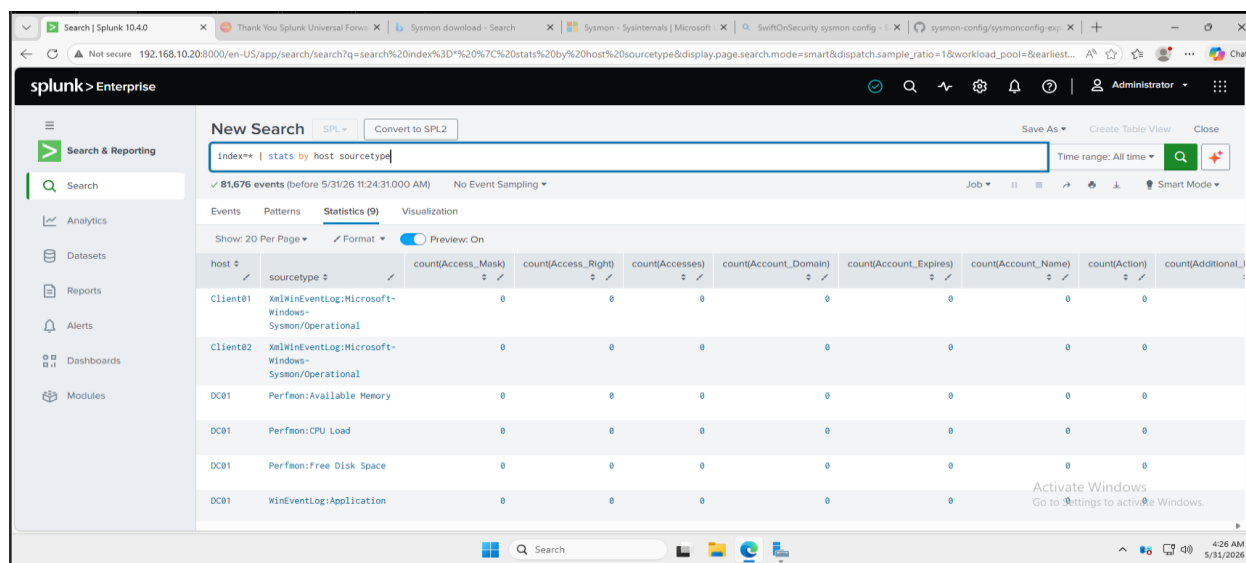
index=* | stats by host sourcetype

Result:

- DC01
- Client01
- Client02

All systems successfully reporting.

Figure 22 – Search History Showing Sysmon Troubleshooting Queries

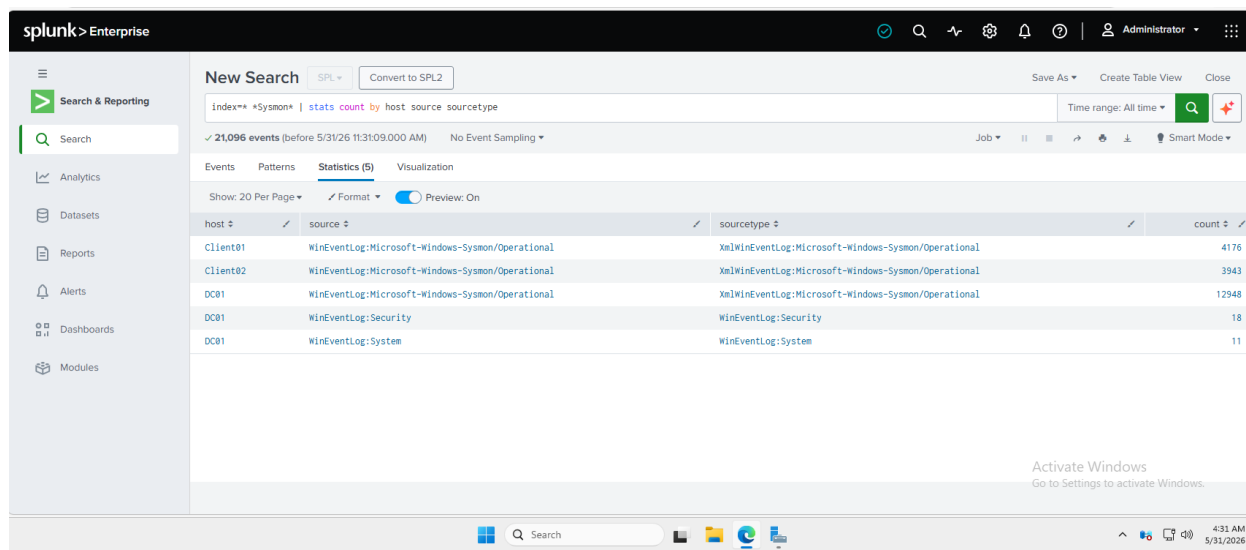


Sysmon Visibility Validation

Search:

index=* *Sysmon* | stats count by host source sourcetype

Figure 23 – Sysmon Host Visibility Validation



Result:

- DC01
- Client01
- Client02

Confirmed Sysmon telemetry ingestion.

Sysmon Event Collection Across Multiple Hosts

Splunk field analysis of Sysmon Operational events showing successful ingestion from three monitored systems: DC01, Client01, and Client02. The results confirm that Sysmon telemetry was collected by the Splunk Universal Forwarder and indexed by Splunk Enterprise for centralized monitoring and threat hunting.

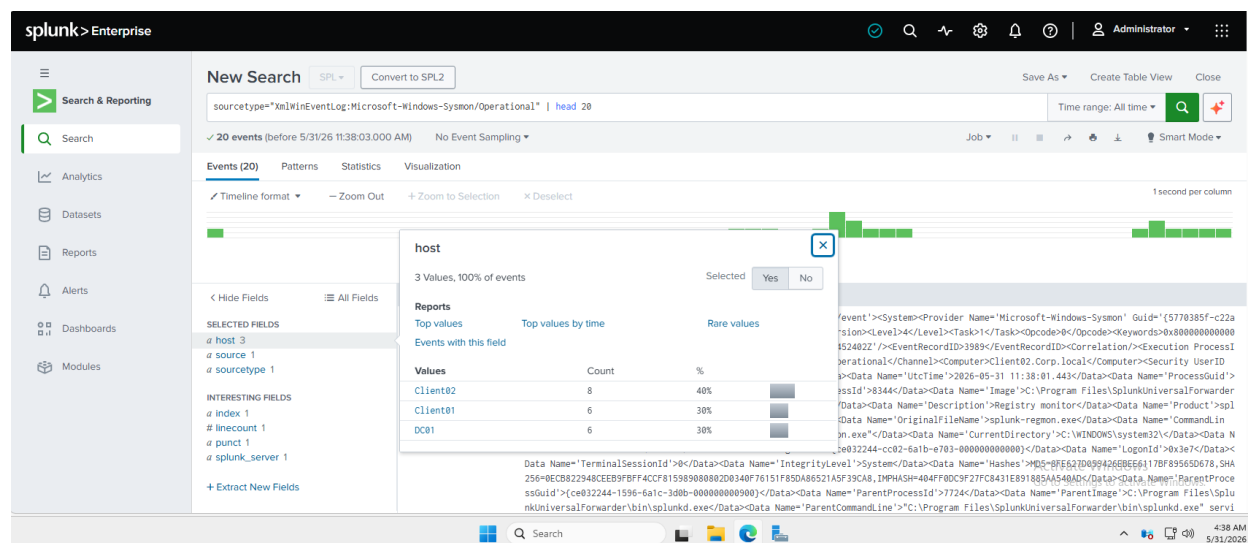
Searched:

`sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" | head 20`

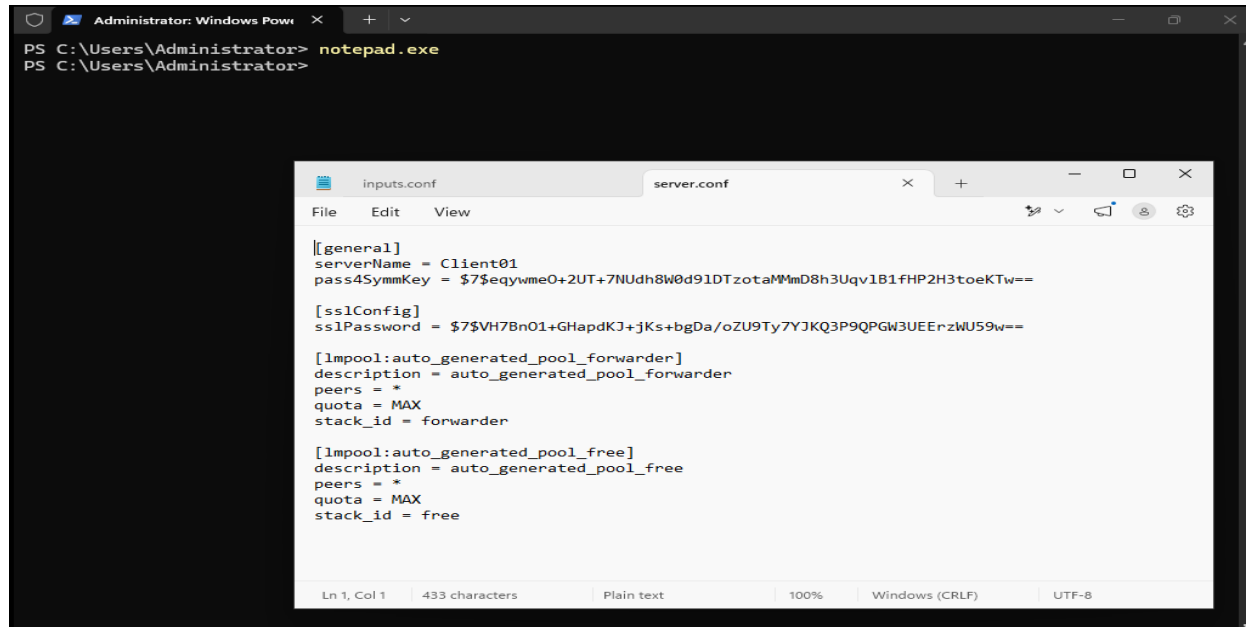
Result:

- Host Client02: Count 8
- Host Client01: Count 6
- Host DC01: Count 6

Figure 24 – Sysmon Event Collection Across Multiple Hosts



Process Creation Validation



Opened:

notepad.exe

on:

- DC01
- Client01
- Client02

Search:

Sysmon Event ID 1

Result:

Notepad process creation events successfully detected.

Figure 25 – Process Creation Validation (Notepad Execution)

New Search [SPL](#) [Convert to SPL2](#) [Save As](#) [Create Table View](#) [Close](#)

`sourcetype='XmlWinEventLog:Microsoft-Windows-Sysmon/Operational' | rex 'EventID:<?<EventID>id>/</EventID>' | rex '<Data Name='Image'>{?<Image>[<+>]}</Data>' | Search EventID=1 Image='*notepad.exe' | table _time host Image | sort _time`

Time range: All time [Search](#)

✓ 22 events (before 5/31/26 12:01:32.000 PM) No Event Sampling [Job](#) [Visualize](#) [Smart Mode](#)

Events Patterns **Statistics (22)** Visualization

Show: 20 Per Page [Format](#) [Preview: On](#) [Prev](#) [1](#) [2](#) [Next](#)

_time	host	Image
2026-05-31 11:58:17.818	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 11:58:17.885	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 10:00:00.909	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 09:24:03.195	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:33:42.300	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:33:42.223	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:31:11.225	Client02	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:31:11.101	Client02	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:30:16.060	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:30:15.926	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:12:26.274	Client02	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe

Activate Windows
Go to Settings to activate Windows.

Figure 26 – Sysmon Event ID 1 Detection Results

New Search [SPL](#) [Convert to SPL2](#) [Save As](#) [Create Table View](#) [Close](#)

`sourcetype='XmlWinEventLog:Microsoft-Windows-Sysmon/Operational' | rex 'EventID:<?<EventID>id>/</EventID>' | rex '<Data Name='Image'>{?<Image>[<+>]}</Data>' | Search EventID=1 Image='*notepad.exe' | table _time host Image | sort _time`

Time range: All time [Search](#)

✓ 22 events (before 5/31/26 12:01:32.000 PM) No Event Sampling [Job](#) [Visualize](#) [Smart Mode](#)

Events Patterns **Statistics (22)** Visualization

Show: 20 Per Page [Format](#) [Preview: On](#) [Prev](#) [1](#) [2](#) [Next](#)

_time	host	Image
2026-05-31 11:58:17.818	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 11:58:17.885	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 10:00:00.909	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 09:24:03.195	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:33:42.300	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:33:42.223	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:31:11.225	Client02	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:31:11.101	Client02	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:30:16.060	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:30:15.926	Client01	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe
2026-05-31 08:12:26.274	Client02	C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad.exe

Activate Windows
Go to Settings to activate Windows.

Event ID Validation

Observed Events:

- Event ID 1
- Event ID 3
- Event ID 11
- Event ID 13
- Event ID 22

Result:

Validated endpoint telemetry collection.

Threat Hunting Activities

Conducted threat hunting using:

- Process Creation Events
- Network Connections
- File Creation Events
- Registry Modifications
- DNS Activity

Figure 27 – Threat Hunting Results Showing Process Activity

_time	host	image	CommandLine	ParentImage
2026-05-31 11:55:41.021	Client01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe		
2026-05-31 11:55:40.765	Client01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe		
2026-05-31 11:55:39.764	Client01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe		
2026-05-31 11:55:39.896	DC01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe		
2026-05-31 11:55:39.006	Client01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe		
2026-05-31 11:55:38.401	DC01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-region.exe		
2026-05-31 11:55:38.156	Client01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorToHandle.exe		
2026-05-31 11:55:37.638	DC01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe		
2026-05-31 11:55:34.384	DC01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe		
2026-05-31 11:55:33.756	DC01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe		
2026-05-31 11:55:32.983	DC01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorToHandle.exe		
2026-05-31 11:55:01.526	Client02	C:\Program Files\SplunkUniversalForwarder\bin\splunk-region.exe		
2026-05-31 11:55:00.417	Client02	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe		
2026-05-31 11:54:59.325	Client02	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe		
2026-05-31 11:54:58.431	Client02	C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe		
2026-05-31 11:54:57.676	Client02	C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe		
2026-05-31 11:54:56.866	Client02	C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorToHandle.exe		
2026-05-31 11:54:42.862	Client01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-region.exe		
2026-05-31 11:54:41.811	Client01	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe		

Examples Detected:

- notepad.exe
- powershell.exe
- cmd.exe
- msedge.exe
- Splunk monitoring processes

Skills Demonstrated

Security Operations

- Security Monitoring

- Threat Hunting
- Event Analysis
- Detection Validation

SIEM Administration

- Splunk Enterprise
- Log Collection
- Indexing
- SPL Queries

Endpoint Monitoring

- Sysmon
- Windows Event Logging

Active Directory

- Domain Services
- DNS
- Domain Management

Windows Administration

- Services
- Event Viewer
- PowerShell
- Troubleshooting

Linux Administration

- Ubuntu Server
- Splunk Deployment

Virtualization

- VMware Workstation Pro
- Multi-VM Infrastructure
- Virtual Networking

Key Accomplishments and Advanced Threat Hunting

Key Accomplishments

Figure 28 – Key Accomplishment Diagram



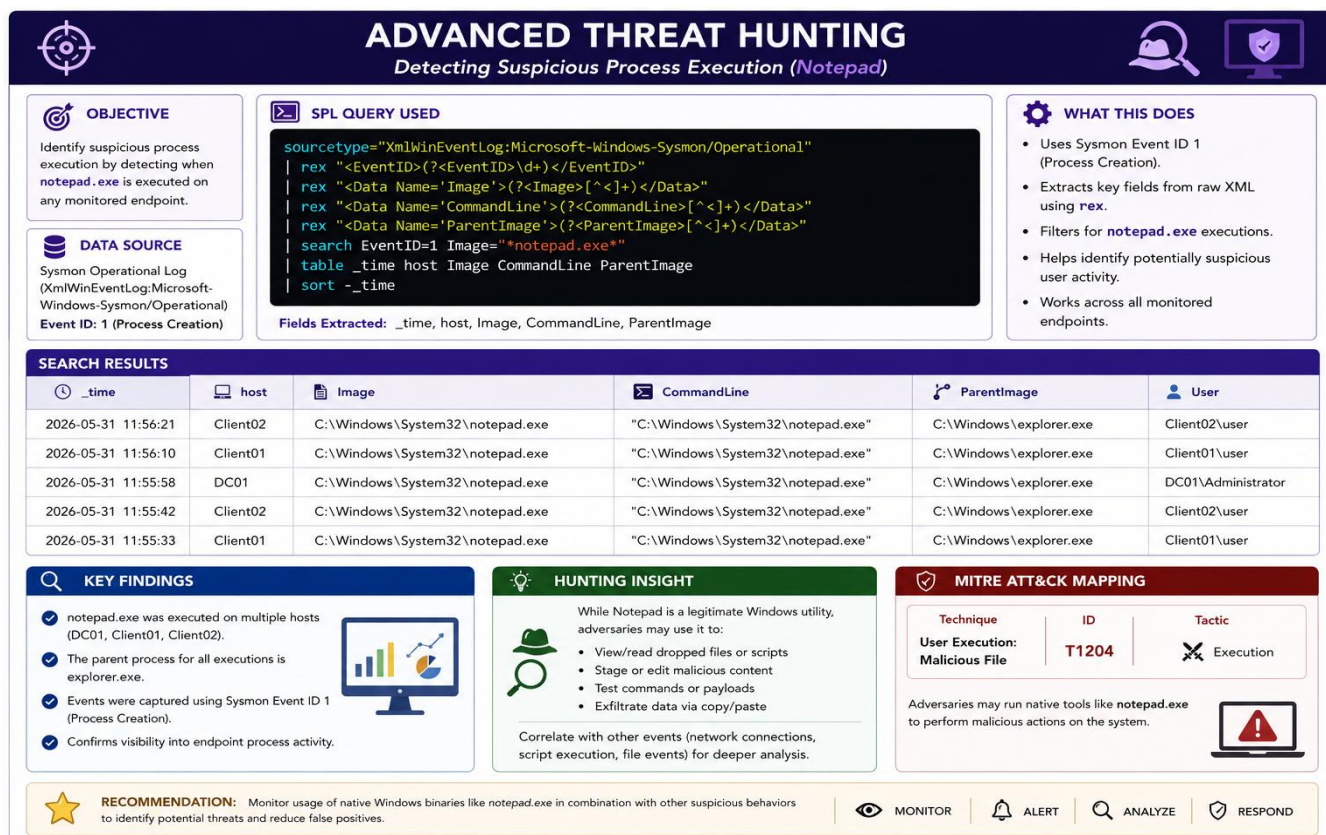
The following objectives were successfully completed during the project:

- Built a multi-host enterprise SOC environment using VMware Workstation Pro.
- Configured Active Directory Domain Services using Windows Server 2025.
- Deployed Splunk Enterprise on Ubuntu Linux.
- Installed Sysmon across multiple Windows systems.
- Configured Splunk Universal Forwarders for centralized log collection.
- Created an end-to-end security monitoring pipeline.
- Troubleshoot Windows Event Log subscription issues and Sysmon visibility problems.
- Validated Sysmon event collection from DC01, Client01, and Client02.
- Successfully collected and analyzed Sysmon Event IDs 1, 3, 11, 13, and 22.
- Performed threat hunting using process creation telemetry.
- Developed custom Splunk SPL searches utilizing XML field extraction.
- Verified end-to-end event flow from endpoint generation through SIEM analysis.
- Demonstrated practical SOC operations and security monitoring workflows.

This project serves as the foundation for future projects involving detection engineering, dashboard creation, alert development, incident response, vulnerability management, Wazuh, Microsoft Sentinel, and advanced threat hunting.

Advanced Threat Hunting Query

Figure 29 – Advanced Threat Hunting Query Diagram



The following custom SPL search was developed to extract process creation telemetry directly from Sysmon XML data:

```
sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
```

```
| rex "<EventID>(?!<EventID>\d+)</EventID>"
```

```
| rex "<Data Name='Image'>(?!<Image>[^<]+)</Data>"
```

```
| rex "<Data Name='CommandLine'>(?!<CommandLine>[^<]+)</Data>"
```

```
| rex "<Data Name='ParentImage'>(?!<ParentImage>[^<]+)</Data>"
```

```
| search EventID=1
```

```
| table _time host Image CommandLine ParentImage
```

```
| sort -_time
```

Caption:

Figure 30 – Advanced Threat Hunting Query Results

This custom Splunk SPL query extracts Sysmon Process Creation events (Event ID 1) from XML-formatted Windows event logs and converts key fields into searchable data. The query uses regular expressions (rex) to parse XML elements and extract information such as the Event ID, executable image, command-line arguments, and parent process. The results are displayed in a table showing when a process executed, on which host it occurred, the executable involved, the command used to launch it, and the parent process responsible for creating it.

This visibility is essential for threat hunting because attackers frequently use legitimate tools such as PowerShell, Command Prompt, Windows Script Host, and other native binaries to execute malicious activity. By analyzing parent-child process relationships and command-line arguments, security analysts can identify suspicious behavior, investigate potential compromises, and validate endpoint telemetry collection.

Query Breakdown

Command	Purpose
<code>sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"</code>	Searches Sysmon Operational logs
<code>rex "<EventID>(?!<EventID>\d+)</EventID>"</code>	Extracts the Sysmon Event ID
<code>rex "<Data Name='Image'>(?!<Image>[^<]+)</Data>"</code>	Extracts the executable path
<code>rex "<Data Name='CommandLine'>(?!<CommandLine>[^<]+)</Data>"</code>	Extracts command-line arguments
<code>rex "<Data Name='ParentImage'>(?!<ParentImage>[^<]+)</Data>"</code>	Extracts the parent process
<code>search EventID=1</code>	Filters for Process Creation events
<code>table _time host Image CommandLine ParentImage</code>	Displays key fields
<code>sort - _time</code>	Sorts newest events first

What the Query Does

Security Value:

- Provides visibility into process execution activity.
- Detects PowerShell, CMD, scripting engines, and administrative tools.
- Reveals parent-child process relationships.
- Assists in identifying malicious or unauthorized software execution.
- Supports incident response and threat hunting investigations.

Threat Hunting Benefits:

- Detects suspicious PowerShell usage.
- Identifies unusual command-line arguments.
- Tracks process execution across multiple hosts.
- Validates Sysmon Event ID 1 collection.
- Demonstrates endpoint telemetry visibility within Splunk Enterprise.

Project Outcome

The Enterprise SOC Lab project was successfully completed and all major objectives were achieved. A fully functional enterprise security monitoring environment was built using VMware Workstation Pro, Windows Server 2025, Windows 11 Pro endpoints, Ubuntu Server, Splunk Enterprise, Sysmon, and Splunk Universal Forwarders.

Sysmon telemetry was successfully collected from multiple endpoints, forwarded through Splunk Universal Forwarders, indexed by Splunk Enterprise, and analyzed through custom SPL threat-hunting queries. The project validated end-to-end visibility across the security monitoring pipeline, from event generation on Windows systems through centralized SIEM analysis.

Throughout the project, multiple operational challenges were identified and resolved, including host visibility issues, Sysmon event collection problems, and XML field extraction limitations. Successful troubleshooting and validation confirmed reliable event collection across DC01, Client01, and Client02.

The completed environment provides a foundation for future SOC analyst activities, detection engineering, dashboard development, alerting, incident response, vulnerability management, Wazuh integration, Microsoft Sentinel integration, and advanced threat hunting exercises.